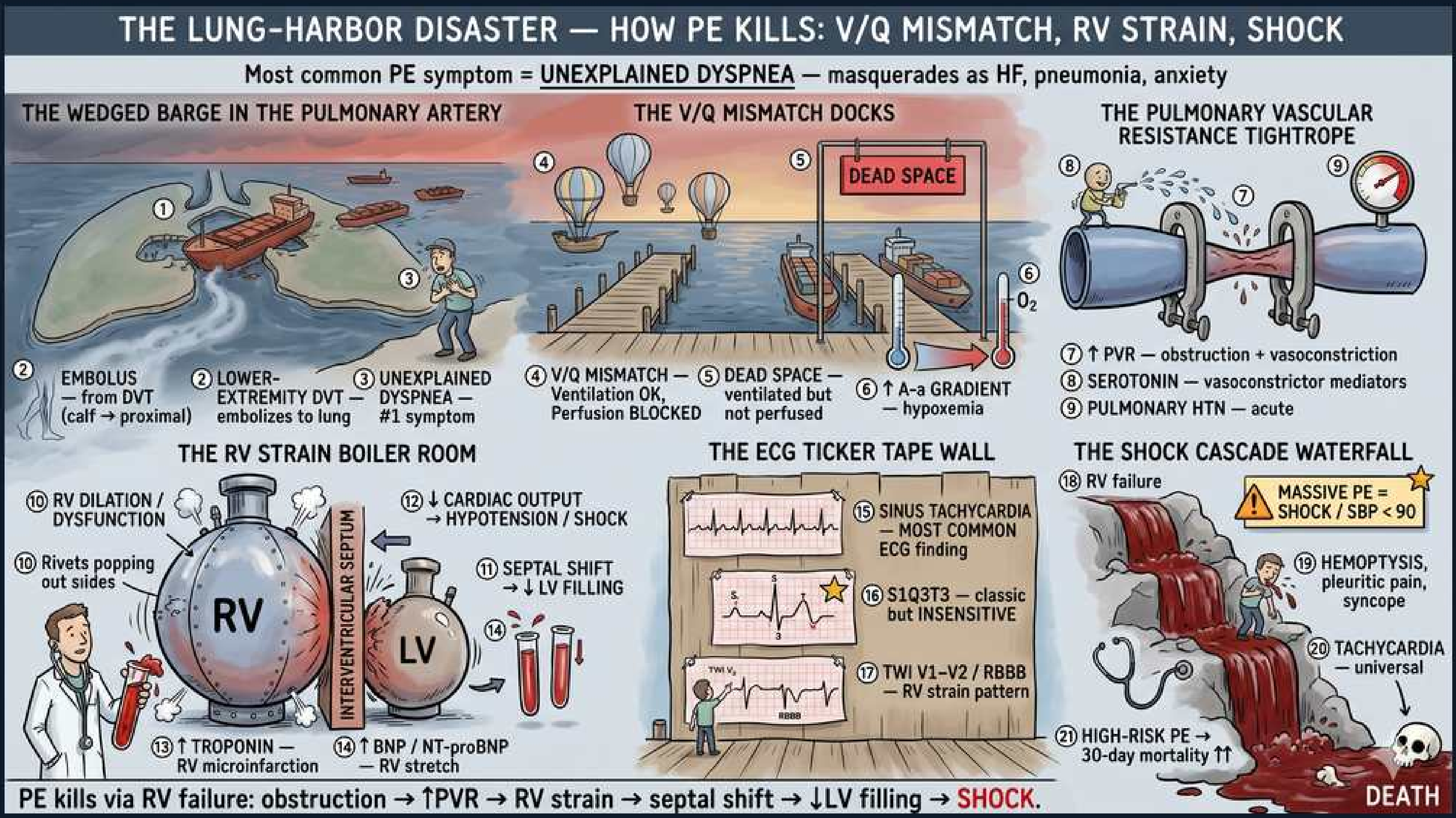
VTE — How PE Kills: V/Q Mismatch, RV Strain & Shock



1. Embolus in pulmonary artery

WHAT YOU SEE

Wedged barge blocking the pulmonary artery channel

WHAT IT ENCODES

Most pulmonary emboli originate from PROXIMAL lower-extremity DVT (popliteal/femoral/iliac), travel through the right heart, and wedge in the pulmonary arteries. A 'saddle' embolus straddles the main pulmonary artery bifurcation and can cause sudden obstructive shock. The hemodynamic impact depends on clot burden plus baseline cardiopulmonary reserve. Mnemonic: 'Leg clot → right heart → lung artery wedge.'

BOARD TRAP

WRONG: 'A saddle embolus by itself mandates immediate systemic thrombolysis regardless of vitals.' Treatment is driven by hemodynamics, not clot location/size on CT — a normotensive patient with a saddle PE is treated as intermediate-risk (anticoagulate, monitor); thrombolysis is for hemodynamic instability.

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2. Unexplained dyspnea

WHAT YOU SEE

Figure gasping, labeled unexplained dyspnea #1 symptom

WHAT IT ENCODES

Sudden unexplained dyspnea is the MOST COMMON symptom of PE, often with pleuritic chest pain and tachypnea; it can masquerade as heart failure, pneumonia, ACS, or anxiety. Syncope or hemoptysis suggests larger/more central PE. A high index of suspicion plus a pretest score (Wells) drives the workup. Mnemonic: 'Sudden breathlessness out of nowhere = think PE.'

BOARD TRAP

WRONG: 'A young, otherwise healthy patient with sudden dyspnea and a clear chest X-ray has anxiety/hyperventilation, so no further workup is needed.' PE classically presents with dyspnea and a normal/near-normal CXR — anchoring on anxiety without applying a pretest score (Wells/PERC) misses a potentially fatal PE.

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3. V/Q mismatch

WHAT YOU SEE

Mismatched dock — ventilation OK but perfusion blocked

WHAT IT ENCODES

PE blocks perfusion to ventilated alveoli, creating ventilation–perfusion (V/Q) mismatch — the dominant mechanism of hypoxemia in PE. Blood is also redistributed to other units, causing relative over-perfusion and additional low-V/Q areas, while tachypnea drives respiratory alkalosis. Mnemonic: 'Air gets in, blood can't — V without Q.'

BOARD TRAP

WRONG: 'PE causes hypoxemia mainly by alveolar hypoventilation with a normal A-a gradient.' PE hypoxemia is from V/Q mismatch (and shunt), which WIDENS the A-a gradient — a hypoventilation pattern (normal A-a gradient, high CO2) points to other causes like opioid overdose or neuromuscular disease.

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4. Dead space

WHAT YOU SEE

Sign labeled DEAD SPACE — ventilated but not perfused

WHAT IT ENCODES

Beyond the occlusion, alveoli are ventilated but not perfused — physiologic dead space — so that ventilation is 'wasted' and does not participate in gas exchange. This raises the dead-space fraction and minute ventilation demand and underlies the elevated end-tidal-to-arterial CO2 gradient sometimes seen in PE. Mnemonic: 'Dead space = ventilation with no perfusion (the mirror of shunt).'

BOARD TRAP

WRONG: 'PE primarily creates a right-to-left shunt (ventilation absent, perfusion present).' PE's hallmark is DEAD SPACE (perfusion absent, ventilation present) — shunt is the opposite physiology (e.g., atelectasis, pneumonia); reversing them is a classic exam discriminator.

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5. Increased A-a gradient

WHAT YOU SEE

Arrow up labeled increased A-a gradient

WHAT IT ENCODES

PE typically widens the alveolar–arterial oxygen gradient due to V/Q mismatch and shunt, often with hypoxemia and a respiratory alkalosis from tachypnea (low PaCO2). However, A-a gradient and oxygen saturation can be normal, especially in young patients with good reserve. Mnemonic: 'Wide A-a gap + respiratory alkalosis = think PE — but normal doesn't exclude it.'

BOARD TRAP

WRONG: 'Normal oxygen saturation and a normal A-a gradient rule out PE.' Up to ~20% of PE patients have a normal A-a gradient and normal SpO2 — absence of hypoxemia does NOT exclude PE; rely on pretest probability, D-dimer, and imaging, not oxygenation.

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6. Increased PVR / vasoconstriction

WHAT YOU SEE

Tightrope/pressure gauge labeled pulmonary vascular resistance

WHAT IT ENCODES

PE raises pulmonary vascular resistance through mechanical obstruction PLUS hypoxic and serotonin/thromboxane-mediated vasoconstriction, producing acute pulmonary hypertension. The previously normal thin-walled RV cannot acutely generate high pressure (it fails above ~40 mmHg mean PA pressure), so afterload rises faster than the RV can compensate. Mnemonic: 'Clot + vasoconstrictors → PVR up → RV can't keep up.'

BOARD TRAP

WRONG: 'A mean pulmonary artery pressure of 70 mmHg in acute PE means the RV is fine because it generated that pressure.' A normal acute RV cannot sustain such pressures — markedly elevated PA pressure in 'acute' PE suggests chronic pulmonary hypertension/CTEPH (an adapted, hypertrophied RV) rather than purely acute disease.

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7. RV dilation / dysfunction

WHAT YOU SEE

Boiler labeled RV swelling/dilation

WHAT IT ENCODES

Acute afterload from PE causes the thin-walled RV to dilate and fail (RV dysfunction), the central event in PE mortality. Imaging/echo signs include RV:LV ratio >1, RV hypokinesis with apical sparing (McConnell's sign), and septal flattening. RV dysfunction with normal blood pressure defines INTERMEDIATE-risk (submassive) PE and prompts close monitoring. Mnemonic: 'Strained RV = the engine that stalls in PE.'

BOARD TRAP

WRONG: 'A normotensive PE patient with echo RV dysfunction (submassive PE) should receive full-dose systemic thrombolysis.' PEITHO showed routine thrombolysis in intermediate-risk PE caused excess major bleeding and stroke without a mortality benefit — anticoagulate and monitor; reserve lysis for hemodynamic decompensation.

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8. Septal shift / decreased LV filling

WHAT YOU SEE

Interventricular septum bowing toward LV

WHAT IT ENCODES

As the RV dilates within the fixed pericardium, the interventricular septum bows toward (and the RV compresses) the LV — ventricular interdependence — reducing LV preload and stroke volume. The result is falling cardiac output and systemic hypotension, the pathway to obstructive shock. Mnemonic: 'Fat RV squeezes skinny LV → low output → shock.'

BOARD TRAP

WRONG: 'Aggressively bolus large volumes of IV fluid in a hypotensive PE patient to raise the blood pressure.' Excess fluid further distends the failing RV, worsening septal bowing and LV filling — give cautious fluids (≤500 mL) and add vasopressors (norepinephrine) and definitive reperfusion; over-resuscitation can precipitate RV collapse.

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9. Troponin elevation

WHAT YOU SEE

Blood tube labeled increased troponin — RV microinfarction

WHAT IT ENCODES

Elevated troponin in PE reflects RV microinfarction/strain from supply-demand mismatch in the pressure-overloaded RV, not coronary plaque rupture. It is a risk-stratification marker: troponin positivity in a normotensive patient (with RV dysfunction) defines intermediate-HIGH-risk PE and predicts worse 30-day outcomes. Mnemonic: 'Troponin up in PE = the RV is hurting → higher risk.'

BOARD TRAP

WRONG: 'Elevated troponin plus dyspnea means acute coronary syndrome — take the patient to the cath lab.' In PE, troponin rises from RV strain, not ACS — anchoring on ACS and giving antiplatelets/cath while missing the PE (and its anticoagulation/reperfusion needs) is a dangerous error; integrate the whole picture (Wells, ECG, imaging).

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10. BNP / NT-proBNP

WHAT YOU SEE

Tube labeled increased BNP/NT-proBNP — RV stretch

WHAT IT ENCODES

BNP/NT-proBNP rise from RV myocardial stretch under acute pressure overload and serve as risk-stratification (not diagnostic) markers in PE: elevation indicates RV dysfunction and intermediate/high-risk disease, while a low level supports low risk and possible outpatient management. Mnemonic: 'Stretched RV leaks BNP = higher-risk PE.'

BOARD TRAP

WRONG: 'A normal BNP rules out PE.' BNP/NT-proBNP are prognostic, not diagnostic — a normal value does not exclude PE (it suggests low risk if PE is confirmed); use D-dimer and imaging to diagnose, BNP/troponin/echo to risk-stratify.

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11. Sinus tachycardia (ECG)

WHAT YOU SEE

ECG ticker tape labeled sinus tachycardia — most common

WHAT IT ENCODES

Sinus tachycardia is the MOST COMMON ECG finding in PE. Other clues include new right bundle branch block, right-axis deviation, T-wave inversions in V1–V3 (RV strain), and atrial arrhythmias. The ECG mainly helps risk-stratify and exclude mimics (e.g., STEMI). Mnemonic: 'PE on ECG? Bet on plain old sinus tach.'

BOARD TRAP

WRONG: 'The classic S1Q3T3 pattern is the most common ECG finding in PE.' S1Q3T3 is the 'classic' but uncommon and insensitive RV-strain pattern — sinus tachycardia is the most common ECG finding; a normal ECG also does not exclude PE.

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12. S1Q3T3

WHAT YOU SEE

ECG segment labeled S1Q3T3 classic but insensitive

WHAT IT ENCODES

S1Q3T3 (prominent S in lead I, Q wave and inverted T in lead III) reflects acute RV strain/cor pulmonale and is the textbook 'classic' PE pattern — but it is uncommon and insensitive, so its absence never rules out PE and its presence is not specific. Mnemonic: 'S1Q3T3 = famous but rare; a footnote, not a screen.'

BOARD TRAP

WRONG: 'No S1Q3T3 on the ECG, so PE is unlikely.' S1Q3T3 appears in a minority of PEs — its absence has no rule-out value; pretest probability, D-dimer, and imaging determine the diagnosis, not this insensitive pattern.

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13. Massive PE → shock

WHAT YOU SEE

Waterfall labeled massive PE → shock, SBP <90

WHAT IT ENCODES

Massive (high-risk) PE is defined by hemodynamic instability: sustained hypotension (SBP <90 mmHg for ≥15 min or requiring vasopressors) or obstructive shock, regardless of clot size on imaging. This is the indication for systemic thrombolysis (e.g., alteplase/tPA 100 mg over 2 h, or 0.6 mg/kg bolus in cardiac arrest), or catheter-directed therapy/surgical embolectomy if lysis is contraindicated or fails. Mnemonic: 'Hypotension = high-risk = give the clot-buster.'

BOARD TRAP

WRONG: 'Hold thrombolytics in a hypotensive massive-PE patient until CTPA confirms the diagnosis.' In hemodynamically unstable suspected PE you don't delay for CTPA — bedside echo showing RV strain supports empiric thrombolysis; waiting for confirmatory imaging in shock risks death.

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14. High-risk PE mortality

WHAT YOU SEE

Skull/death sign labeled high-risk PE 30-day mortality

WHAT IT ENCODES

High-risk (massive) PE carries high short-term mortality (often >15–25% at 30 days) driven by obstructive shock and acute RV failure. Management combines reperfusion (systemic thrombolysis, or catheter-directed/surgical embolectomy if lysis is contraindicated or fails) with vasopressor (norepinephrine) and cautious volume support; ECMO is a bridge in refractory collapse. Mnemonic: 'High-risk PE = RV failure + shock = race to reperfuse.'

BOARD TRAP

WRONG: 'Treat a high-risk PE in shock with anticoagulation (heparin) alone and admit for monitoring.' Anticoagulation prevents new clot but does not rapidly relieve the existing obstruction — high-risk/massive PE requires reperfusion (thrombolysis or embolectomy) plus hemodynamic support; heparin alone is insufficient.